TASK DELIVERABLE #8

Literature Review

Smart Emergency Response & Healthcare Coordination Platform

1. Existing emergency healthcare systems

In Egypt you call **123** when you need an ambulance. In 2022 the Ministry of Health announced **112**, one number for ambulance, police and fire, and said it would detect the caller's location automatically [1]. The same report said ambulances take around 8 to 15 minutes to arrive on average [1]. We only found this number in that 2022 article, so it may be different now.

Still, most of the process happens over the phone. The caller has to explain what happened and where they are, which is hard when you are panicking. The hospital usually knows nothing about the patient until the ambulance arrives, and the family often hears about it late. WHO's guide for ambulance systems spends a lot of time on dispatch and on the handover between the ambulance and the hospital [2]. That handover is the part our project focuses on.

2. Existing applications and solutions

We looked at some systems and apps that already help in emergencies:

Solution	What it does	What is missing for us
123 / 112 hotlines (Egypt)	You call and talk to the emergency service. 112 is supposed to get your location [1].	No medical profile and no live status. The hospital isn't told before arrival.
Apple Medical ID	Keeps your allergies, medications and emergency contacts on the iPhone so responders can see them [3].	Stays on the phone. The dispatcher and the hospital don't get it.
Android Emergency Location Service	Sends your location when you call an emergency number [4].	Just the location, nothing after the call.
RapidSOS (USA)	Sends data from phones and devices to 911 centers [5].	Made for 911 in the US. Not used in Egypt.
Hospital systems	Manage beds, departments and patient files inside the hospital.	Not connected to ambulances or the emergency center.

3. Relevant technologies and approaches

This is a Full Stack .NET project, so we are building it as a website. The backend uses ASP.NET Core Web API with Entity Framework Core and SQL Server, and the frontend uses React. Patients and ambulance crews will mostly open it from their phones, so the pages have to be responsive.

We will use SignalR for live updates. When the ambulance changes its status, everyone sees it without refreshing. The patient's location comes from the browser's GPS and shows up on a map.

The system stores medical data, so we are using ASP.NET Core Identity with JWT and role-based access. A hospital user, for example, shouldn't see the same things as an admin.

There will also be AI-assisted triage that suggests the emergency type, priority and department, plus simple matching rules that suggest an ambulance and a hospital. Staff still make the final call in both.

4. Comparison with our project

Step	Today (by phone)	With our platform
Asking for help	Call and explain everything.	Press the emergency button and pick the type.
Location	The caller describes it.	GPS location goes with the request.
Medical info	Only what the patient can say.	Staff can open the Emergency Medical Profile.
Choosing ambulance / hospital	Done by hand.	The system suggests, staff confirm.
Hospital getting ready	After the patient arrives.	Before arrival, with patient info and ETA.
Family	They keep calling to ask.	Emergency contacts get notifications.
History	Paper, or different systems.	Every status change is saved.

5. Strengths and limitations of existing solutions

The current solutions do some things well. People know the hotline numbers, and they work from any phone even without internet. Medical ID and emergency location already help responders, and ambulance teams have clear procedures to follow [2].

The weak point is that every part works on its own. The phone, the dispatcher, the ambulance and the hospital don't see the same information. Since almost everything goes through phone calls, details get lost when people are stressed, and the hospital and family end up finding out late.

6. Identified gaps

Gap	What our project does about it
No one place to follow the case	Dashboards for the emergency center, ambulance, hospital and doctor, all looking at the same case.
Medical info isn't there when it's needed	An Emergency Medical Profile that only authorized staff can see.
Hospital isn't ready	The hospital gets the patient's info and ETA before arrival.
Family doesn't know what's going on	Notifications to emergency contacts.
Ambulance and hospital picked by hand	Matching suggestions that staff confirm.

7. Feedback and evaluation

We will fill in the lecturer's feedback here after the review.

Date	Lecturer comment	What we did

Before submitting, we also read each other's documents (the Proposal, Project Plan and KPIs) and found a few things that didn't match. The Proposal had the Emergency Medical Profile and emergency contacts,

but the Plan didn't, so we added them to the Plan. We also made sure every file says the system is a website and that the ambulance and hospital are suggested, not picked automatically.

8. Suggested improvements

These depend on how much time we have after the MVP. An Arabic version of the website would help a lot of users. Sending SMS to the family would cover the cases where they have no internet. The AI triage could get better with more examples, but it would stay a suggestion. Hospitals could also update their free capacity themselves, which would make hospital matching more accurate. Connecting with 123 or 112 would be great one day, but that needs official access we don't have.

9. Final grading criteria

The lecturer sets the weights. This is what we plan to deliver for each part.

Part	What is evaluated	What we will show	Weight
Documentation	Planning, requirements, analysis and design	Proposal, Plan, KPIs, diagrams, README	By lecturer
Implementation	Working features, code quality, GitHub work	Source code, pull requests, deployed website	By lecturer
Testing	Test cases, bug fixes, KPI results	Test cases, bug reports, KPI results	By lecturer
Presentation	Demo and explaining our work	Final presentation and live demo	By lecturer

References

1. Egyptian Streets (2022). *112: Egypt Announces Its Very Own '911' Hotline for Public Emergencies.* egyptianstreets.com
2. World Health Organization (2025). *Prehospital emergency care: operational guidance for ambulance systems.* who.int
3. Apple Support. *Set up your Medical ID in the Health app on your iPhone.* support.apple.com/105072
4. Android. *Emergency Location Service: How it works.* android.com
5. Business Wire (2022). *Comtech Partners with RapidSOS to Provide Life Saving Data to 911 Agencies Across the United States.* businesswire.com